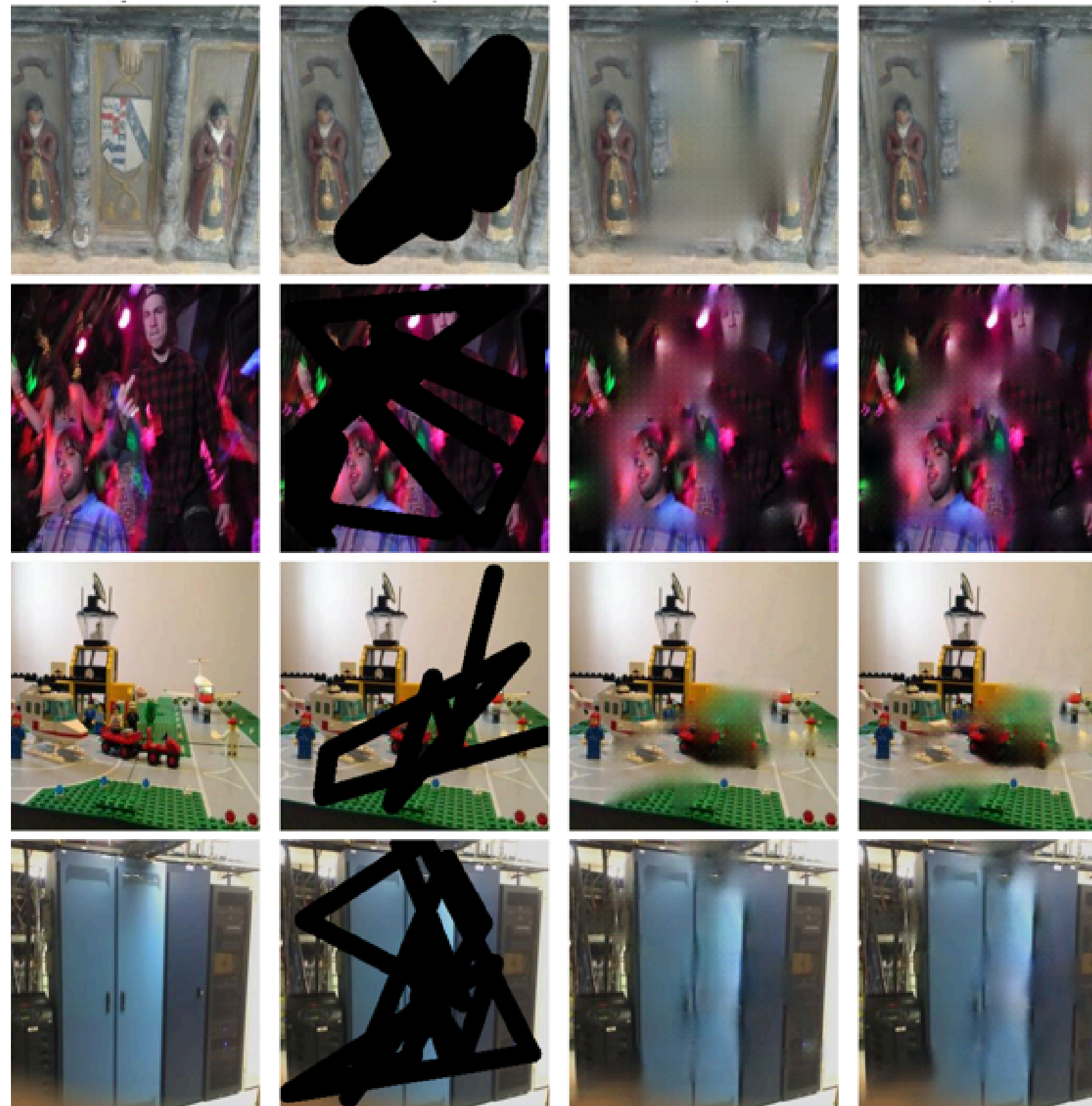


Group 4

Boundary-Aware Semantic Image Inpainting with Spectral Coherence Supervision

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What We'll Cover

Motivation and Problem

Our Approach & Architecture

Loss Function Design

Experiments and Results

Discussion & Conclusion



Why Does Inpainting Boundary Quality Matter?



Object removal in
photography,
media restoration

Existing methods
optimise globally

Spectral discontinuity at
boundaries– artifacts
invisible to pixel-level losses
but obvious to the human eye

The Inpainting Problem

Input: RGB image + binary mask (10–60% coverage, free-form strokes)



Output: Plausible, photo-realistic completion of the masked region



Core challenge: Standard losses (L1, perceptual) treat all pixels equally — boundary pixels need special treatment

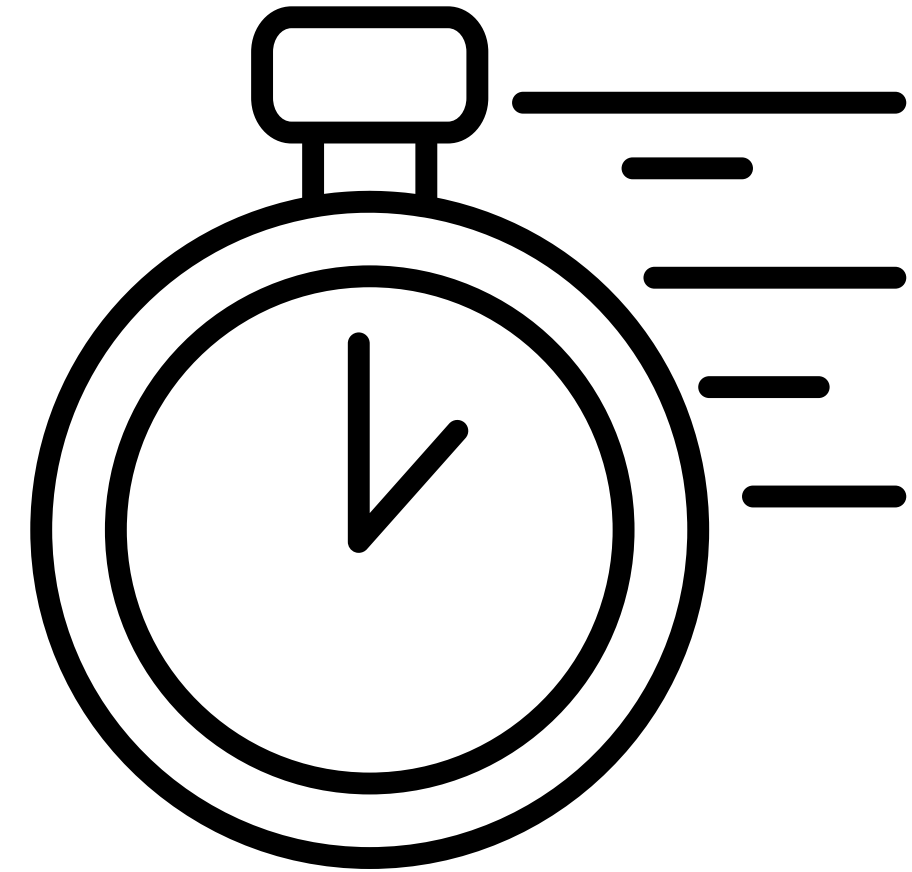


Our focus: The boundary band — a narrow zone around the mask edge where generation meets reality



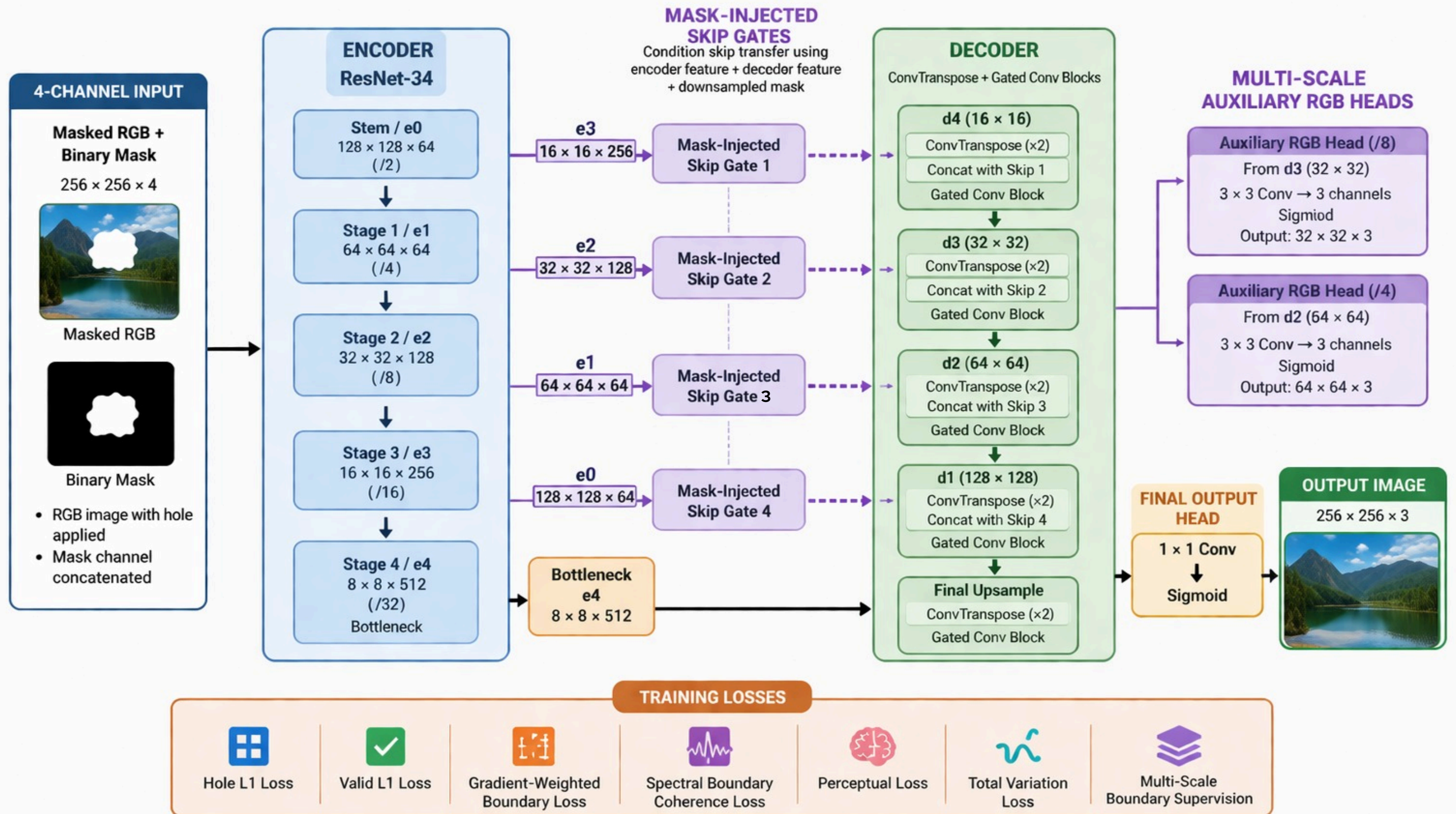
Prior Work & The Gap

Method	Contribution	Limitation
PConv (Liu et al., 2018)	Partial convolutions for masked regions	No boundary-specific supervision
DeepFillv2 (Yu et al., 2019)	Gated convolutions, free-form masks	Perceptual loss only
LaMa (Suvorov et al., 2022)	Fourier convolutions, large receptive field	No localised spectral boundary term



No prior work applies frequency-domain supervision localised to the boundary band. Spectral coherence at boundaries remains unaddressed.

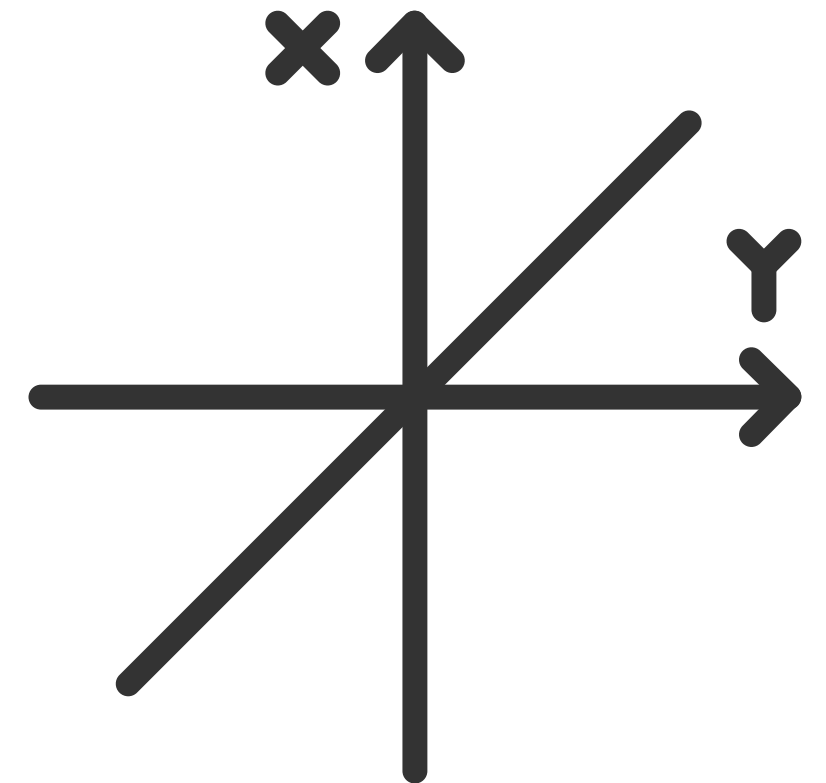
Model Architecture



Our Loss Function

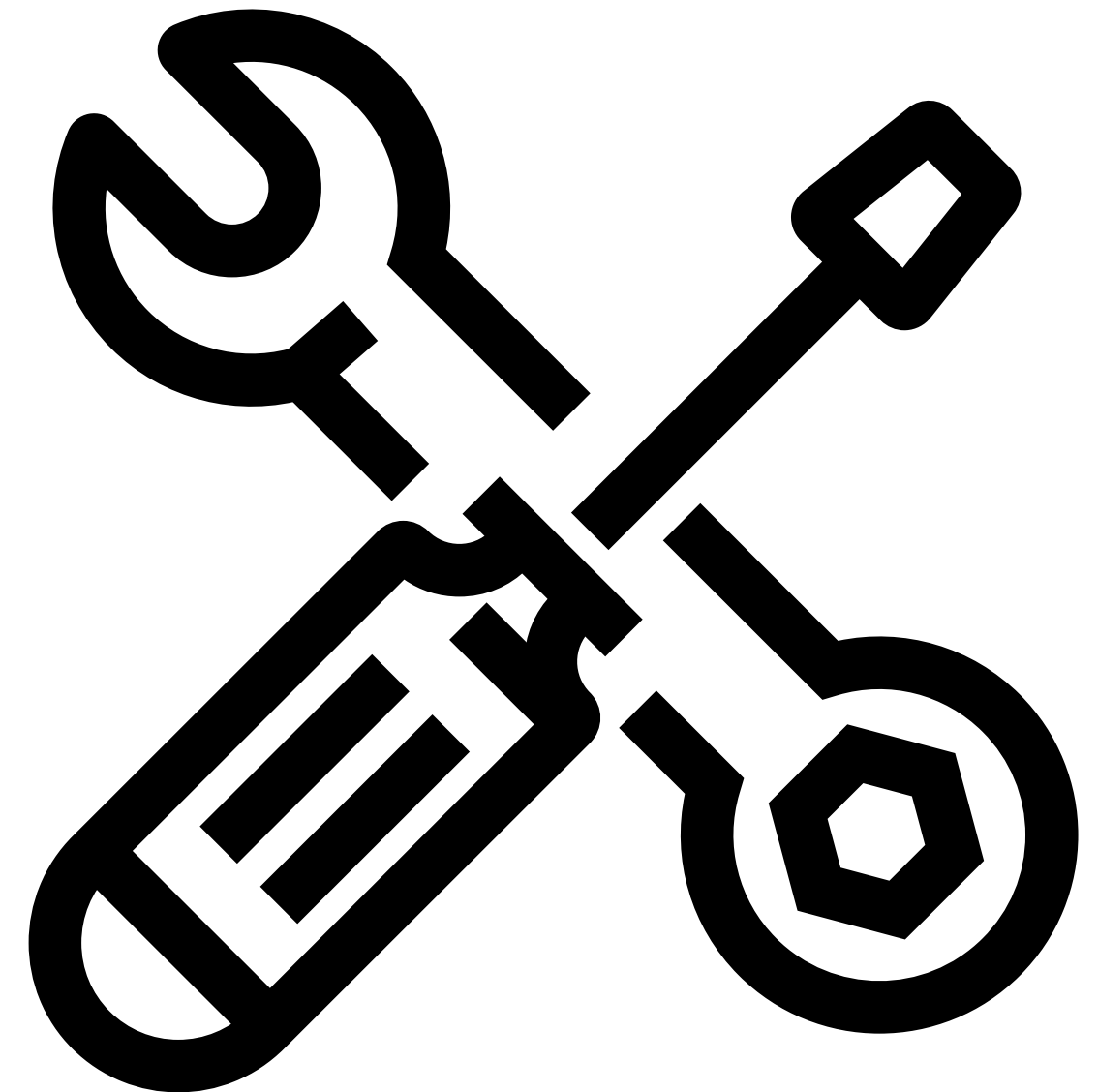
- Hole + Valid L1 — pixel reconstruction inside and outside the mask
- Perceptual (VGG16) — semantic feature matching up to relu3_2
- Boundary L1 — gradient-weighted L1 restricted to the boundary band
- Spectral Boundary Coherence — FFT magnitude mismatch on boundary patches
- Total Variation — smoothness inside the hole only

$$\begin{aligned}\mathcal{L}_{total} = & \lambda_h \mathcal{L}_{hole} + \lambda_v \mathcal{L}_{valid} + \lambda_p \mathcal{L}_{perceptual} \\ & + \lambda_b \mathcal{L}_{boundary} + \lambda_s \mathcal{L}_{spectral} \\ & + \lambda_{tv} \mathcal{L}_{TV} + \lambda_{ms} \mathcal{L}_{multi_scale}\end{aligned}$$



Setup

- Dataset: Places365 (31,971 images), 70/15/15 split
- Cross-domain: CelebA-HQ (30,000 faces), zero-shot
- Hardware: Tesla T4 (15.6 GB VRAM), Kaggle
- 4-phase progressive ablation (backbone → skip → conv → loss)



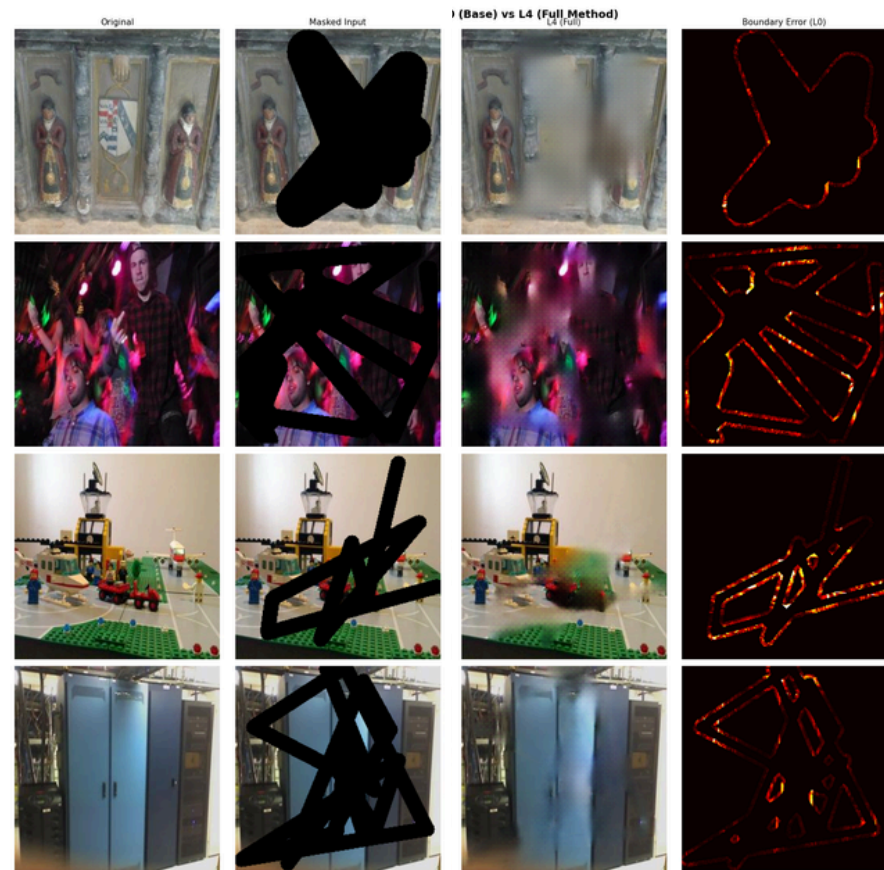
Experiments

Experiment	PSNR $\uparrow$	SSIM $\uparrow$	Hole-MAE $\downarrow$	Boundary-MAE $\downarrow$	Spec. Coh. $\uparrow$
L0 (Base)	20.37	0.7150	0.1034	0.0319	0.9839
L1 (Uni. Bound)	20.09	0.7099	0.1076	0.0320	0.9833
L2 (Grad Bound)	20.07	0.7081	0.1081	0.0330	0.9831
L3 (Spectral Only)	20.08	0.7118	0.1073	0.0320	0.9834
L4 (Full Method)	20.02	0.7132	0.1078	0.0311	0.9836

Our method targets boundary quality rather than overall reconstruction accuracy.

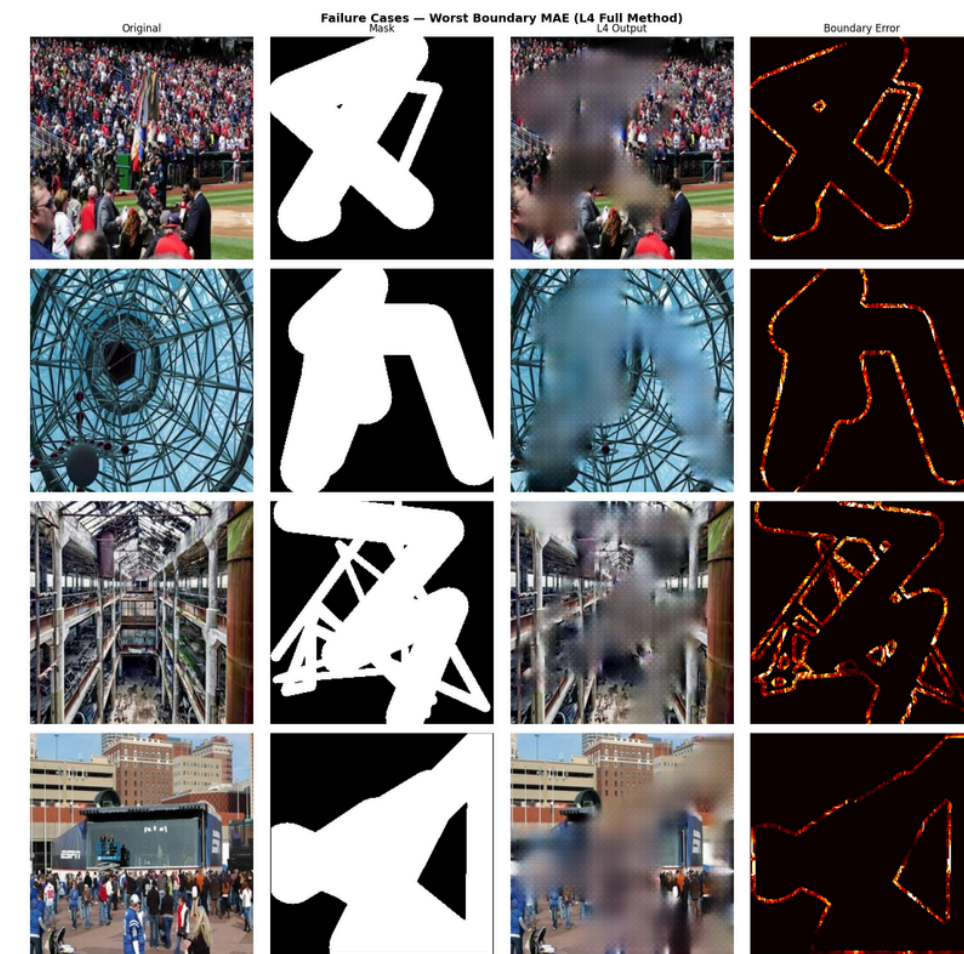


Results: What Works and Where We Fail



Success

Organic textures and natural scenes:
L4 produces cleaner boundary transitions



Failure

Large masks >50% over structured content:
model hallucinates implausible structures

Summary & Future Directions



Contributions:

Introduced

SpectralBoundaryCoherenceLoss – first
FFT-based supervision localised to the
inpainting boundary band

Demonstrated that Boundary-MAE
improves consistently across all mask
difficulty levels with full method

Conducted a rigorous 4-phase ablation
isolating each architectural and loss
component's contribution



Future Work:

Add adversarial (GAN) training to recover
global PSNR while retaining boundary gains

Fine-tune on domain-specific datasets
(faces, text, satellite) to improve cross-
domain transfer

Extend boundary supervision to diffusion-
based and Transformer inpainting
backbones



Thank you

We're open to questions!

